ANANYAA SRINIVASAN

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EDUCATION

University of Pennsylvania, School of Engineering and Applied Sciences — **Philadelphia, PA**
Masters in Biotechnology (Concentration: Molecular Biology) | GPA: 3.95/4.00 — May 2025
Vellore Institute of Technology, School of Biosciences and Technology — **Vellore, India**
Bachelor of Technology (Major: Biotechnology) | GPA: 3.96/4.00 — June 2023

SKILLS

Software: JMP, GraphPad Prism, SnapGene, EPIC, FlowJo, ELN, FCS Express, SpectroFlo, Design of Experiments (DOE), Python, Microsoft Office
Laboratory: PCR, mammalian cell culture, molecular cloning, bacterial transformation, plasmid DNA prep, DNA isolation (mini and maxi-prep), transfection, lentiviral packaging, transduction, ELISA, DLS, gel electrophoresis, multi-color and spectral flow cytometry (Fortessa, A3 Lite, Apogee, Lyric, Cytek Aurora), spectral unmixing, flow cytometry panel design, automated liquid handlers (Hamilton Tecan), fluorescence in-situ hybridization

PROFESSIONAL EXPERIENCE

Department of Pathology, Stanford School of Medicine — **Palo Alto, CA**
Life Science Research Professional — September 2025 – Present
- Established and operationalized a functional immunology laboratory, overseeing assay development and lab management.
- Led end-to-end development, optimization and clinical translation of diagnostic assays for functional immune disorders and pathogen-driven disease, including a DHR-based neutrophil oxidative burst assay and a flow-FISH assay for EBV detection.
- Liaised directly with Stanford hospital clinical laboratories to transfer and implement novel assays into routine patient specimen testing workflows, enabling faster and more practical diagnostic alternatives to existing methods.
- Identified and banked clinically relevant patient specimens; established cryopreserved repositories of whole blood, bone marrow, primary immune cells, and reference cell lines to support assay validation.
- Onboarded institutional spectral flow cytometer (Cytek Aurora) for Stanford Flow Cytometry Clinical Laboratory; leading assay optimization and CLS training on the platform.

THINK, Carl June Lab, Centre for Cellular Immunotherapies — **Philadelphia, PA**
Research Assistant — September 2023 – June 2025
- Advanced preclinical development of engineered immune cell therapies by isolating, activating and co-culturing immune cells from PBMCs with feeder cells and designing functional assays to improve cytotoxicity against tumor models.
- Cultured, expanded and maintained diverse cell types (primary immune cells, 293T, feeder, and cancer cell lines) for high-throughput experimental validation and quality control.
- Performed molecular cloning for cell engineering, including vector design, PCR, Gibson and Golden Gate assembly, bacterial transformation and gene sequencing.
- Produced and concentrated high-titer lentivirus via 293T transfection and ultracentrifugation, followed by transduction of mammalian cell lines for stable gene integration.

Moderna Therapeutics — **Norwood, MA**
Co-op, LNP Process Development — July 2024 – December 2024
- Addressed the challenge of mixer scalability in mRNA-LNP formulation by leading a scale-up assessment comparing particle size and PDI across flow rates to identify a mixer with consistent performance from small to large scale conditions.
- Executed end-to-end mRNA-LNP production across 7+ projects, optimizing formulation parameters, including process flow, ingredients and their ratios, and neutralization buffers, to enhance nanoparticle quality relative to the standard process
- Applied tangential flow filtration, ultrafiltration, and diafiltration and utilized automated liquid handlers for sample preparation for nanoparticle characterization using dynamic light scattering and flow cytometry.
- Ensured process safety by monitoring real-time system pressure with PendoTECH sensors across all stages of production.

ACADEMIC PROJECTS

Study of ICAM – 1, aCD56, 41BBL and CD58 for CD2 stimulation in natural killer (NK) cells — **July 2025**
- Engineered feeder cell systems via molecular cloning and lentiviral transduction to investigate the impact of ICAM-1, CD58, 41BBL and aCD56 feeder cell systems on NK cell expansion, viability and cytotoxicity against Raji and U251 tumor cells.
- Independently developed an AI-based predictive model for NK cell cytotoxicity across feeder configurations and tumor targets, trained on experimental cytotoxicity data from Raji and U251 cell lines.

Zika virus vaccine formulation — **December 2023**
- Designed an mRNA Zika virus vaccine, optimizing antigen expression by human codon optimization and incorporating genetic modifications, including specific mutations in E-DII-FL and use of mosquito salivary protein.
- Proposed animal studies for pre-clinical validation, clinical trial strategies, and a pipeline driven development approach.

PUBLICATIONS

Immunization against COVID-19: A Comprehensive Review on the Leading Vaccines — **January 2021**
- Conducted a comprehensive review of key vaccines in use including production, trial data, efficacy, and safety.
- Published on 'Coronaviruses' (https://doi.org/10.2174/0126667975285709231219080802)